Investigations, Procedures and the Perioperative Record

Critical results, operative notes, implant traceability and the nine documents a single operation must generate.

Module	07 of 12
Study time	105 minutes
Audience	Surgeons, anaesthetists, theatre nursing, laboratory and imaging staff, medical records staff
Assessment	Knowledge check plus a 50-question scored bank
Standards basis	NABH 6th Edition COP chapter on surgical and anaesthesia care; AAC chapter on diagnostic services; WHO Surgical Safety Checklist

A single operation generates nine documents, each with a different author and a different deadline, and the file is complete only when all nine are present. Investigations generate their own documentation duty, most sharply where a result is critical and somebody must be shown to have acted on it. This module covers the result pathway, the perioperative trail, the operative note, implant traceability and the procedures performed outside theatre that are routinely under-documented.

LEARNING OUTCOMES — by the end of this module you will be able to:

- Document a critical result from identification through to the clinical action taken
- Name the nine perioperative documents, their authors and their deadlines
- Write an operative note containing every required element
- Record implant and prosthesis details so that a recall can be answered within a working day
- Document anaesthesia, sedation and post-anaesthesia recovery to the required standard
- Document procedures performed outside the operating theatre to the same standard as those inside it
- Document specimens so that the chain from patient to report is unbroken

7.1 Ordering an investigation

The requisition is part of the record. An order with no clinical details produces a report without clinical interpretation, and a laboratory or radiologist who cannot see the question cannot answer it. The requisition must carry full patient identifiers, the test requested, the relevant clinical information, the requesting clinician identified, and the date and time.

- Record in the clinical notes why the investigation is being ordered — the question it is meant to answer
- Where a test is urgent, record that it was marked urgent and the time it was sent
- Record relevant details the laboratory needs: current antibiotics for a culture, last menstrual period for imaging, renal function before contrast, anticoagulants before a procedure
- Label every specimen at the bedside, after positive patient identification, by the person who collected it
- Record the time of collection, because for many tests the result cannot be interpreted without it

CAUTION
Pre-labelled specimen containers and containers labelled at the nursing station are the two mechanisms by which a result is attributed to the wrong patient. Both are documentation failures and both can be fatal, particularly for cross-match samples.

7.2 Receiving and acting on results

A result that arrives and is filed without being read is worse than a result that was never ordered, because the hospital now holds the information and did nothing with it. The record must show that results were seen and considered, and this is done by referring to them in the clinical notes, not by their mere presence in the file.

Critical results

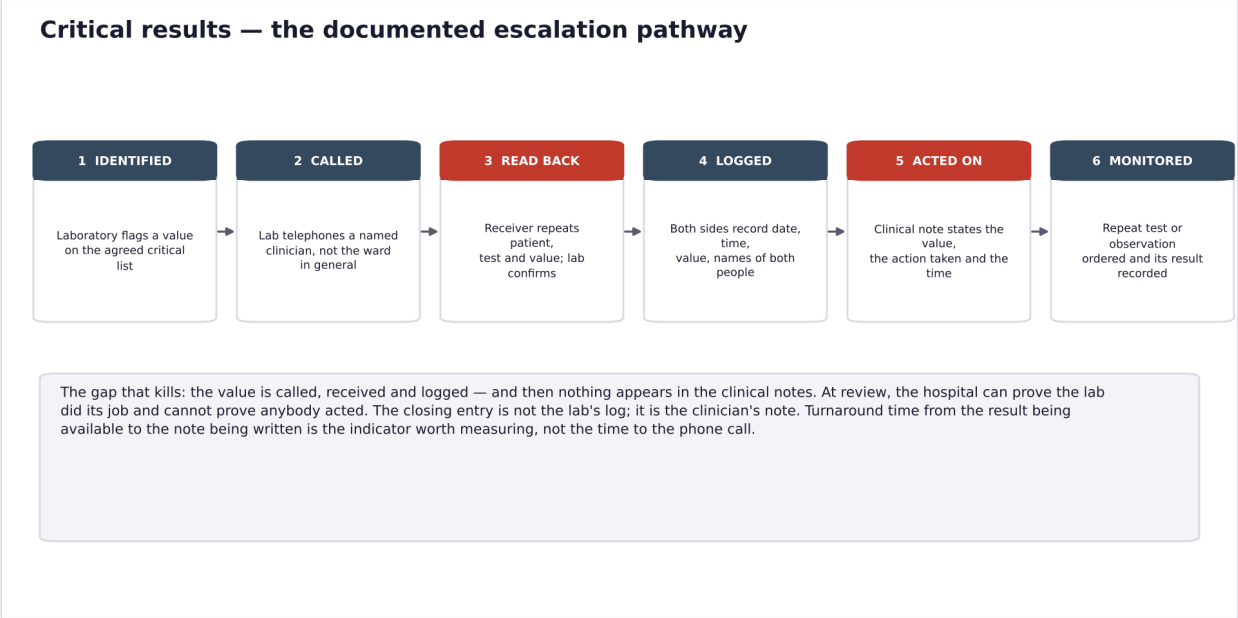


Figure 7.1 — The documented escalation pathway for a critical result.

Every hospital must have a defined list of critical values, a defined time within which they must be communicated, and a defined read-back procedure. The chain has six steps and the last one is the one that fails.

The laboratory identifies the value, telephones a named clinician, obtains a read-back, and logs the call with both names and the time. At that point the laboratory's duty is discharged and the clinician's begins. What is then needed is an entry in the clinical record stating the value, the time it was received, the action taken, and the follow-up arranged. Audits repeatedly find the first four steps complete and the fifth absent, which means the hospital can prove the result was communicated and cannot prove anyone acted.

> **KEY POINT**
> The indicator worth measuring is not the time from result to phone call. It is the time from result to the clinical note recording the action taken. The first measures the laboratory; the second measures the hospital.

7.3 The perioperative paper trail

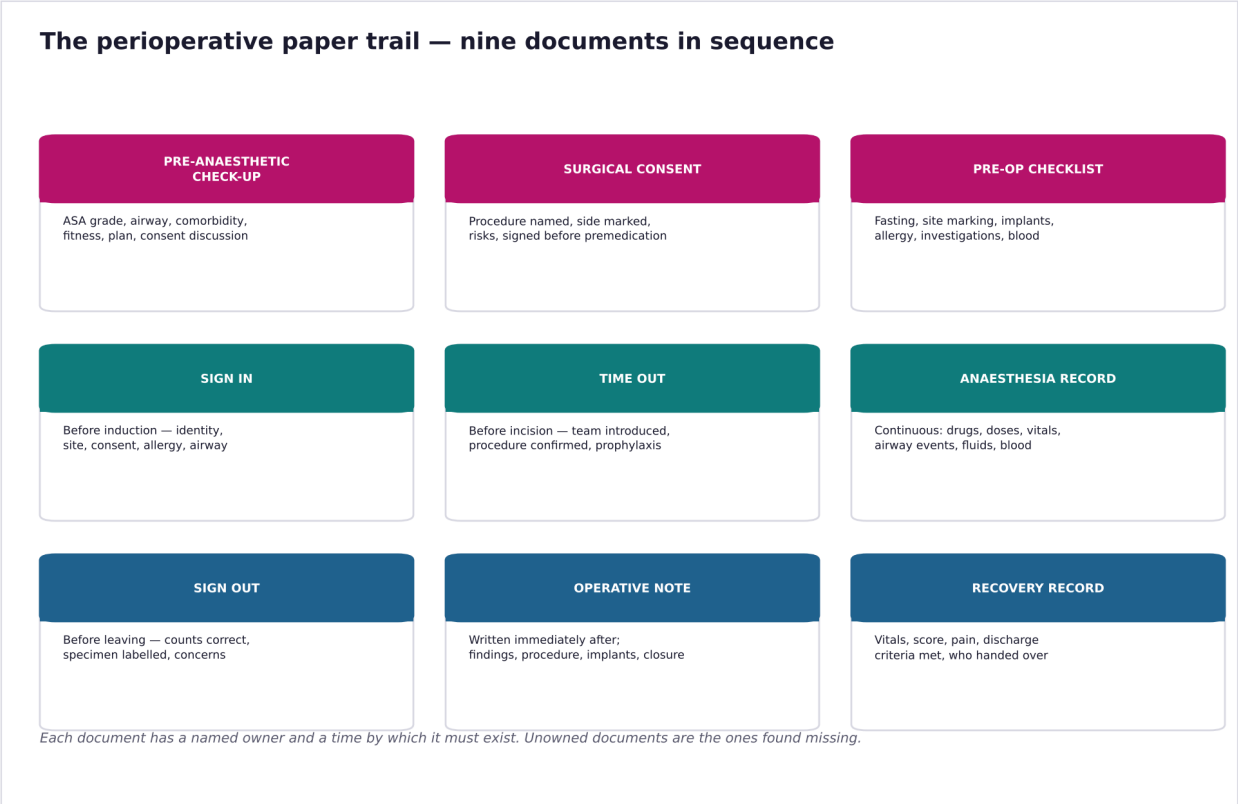


Figure 7.2 — The nine documents a single operation generates.

Each of the nine has a named owner and a deadline. Documents without an owner are the ones found missing, and the two most frequently missing are the pre-anaesthetic assessment for emergency cases and the recovery record for cases that return to the ward quickly.

Document	Author	Deadline	What is most often missing
Pre-anaesthetic check-up	Anaesthetist	Before the patient enters theatre; for electives, in advance of the admission	ASA grade, airway assessment, fasting status, and the plan discussed with the patient
Procedure-specific consent	The operating surgeon or a clinician able to perform the procedure	Before premedication	Side and site in words, risks completed, registration number of the explainer
Pre-operative checklist	Ward nursing, verified in theatre reception	Before transfer to theatre	Implant availability, blood arrangements, allergy confirmation, removal of prostheses
Sign in	Anaesthetist with theatre nurse	Before induction	Confirmation of consent and site marking with the patient awake
Time out	Whole team, led by a nominated member	Before incision	Team introductions, antibiotic prophylaxis timing, anticipated critical events
Anaesthesia record	Anaesthetist	Continuously during the procedure	Drug doses and times, airway events, fluids and blood, and the end-of-procedure handover
Sign out	Theatre nurse with the team	Before any team member leaves	Instrument, sponge and sharps counts, specimen labelling confirmation, equipment problems

Document	Author	Deadline	What is most often missing
Operative note	Operating surgeon	Immediately after the procedure, before leaving the theatre suite	Findings, implants, estimated blood loss, complications, and post-operative instructions
Post-anaesthesia recovery record	Recovery nursing with the anaesthetist	From arrival in recovery to discharge from it	Discharge criteria met with the time, and the name of the person to whom the patient was handed over

7.4 The operative note

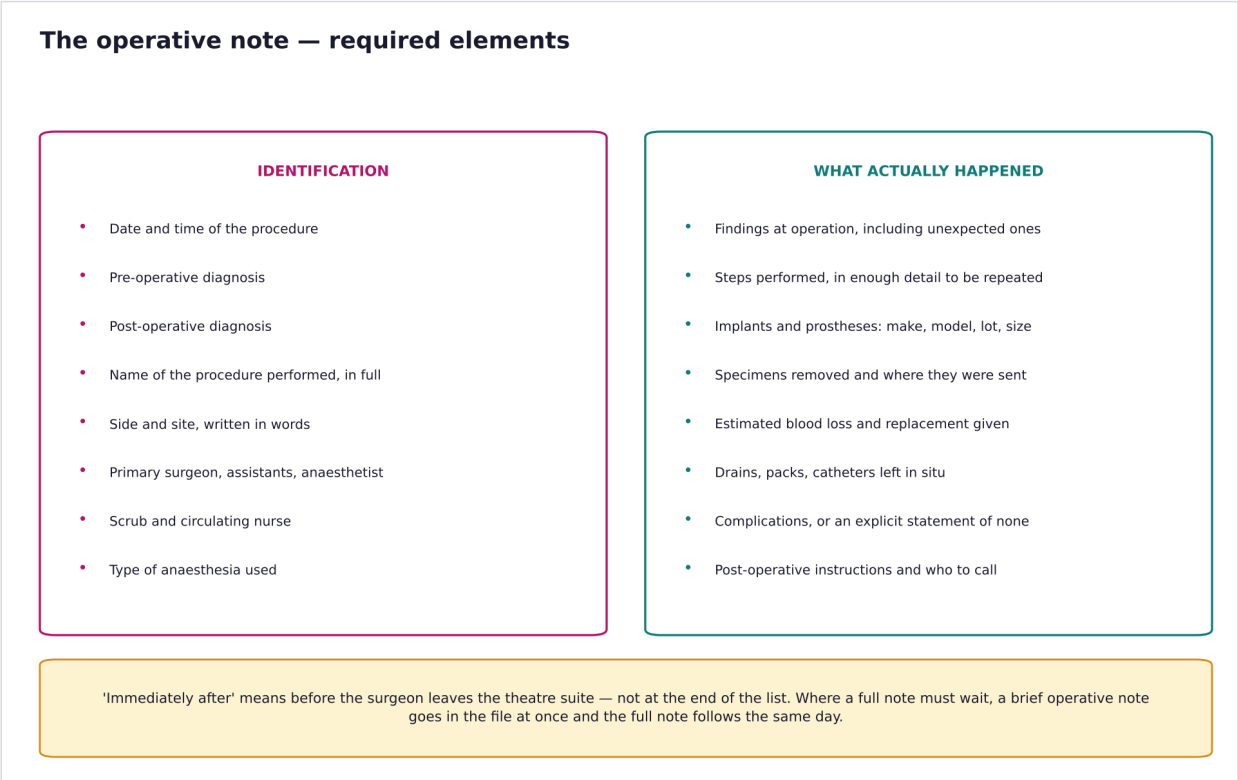


Figure 7.3 — Required elements of an operative note.

“Immediately after the procedure” means before the surgeon leaves the theatre suite, not at the end of the list. Notes written for four patients at the end of a list are written from memory, and the details that matter — estimated blood loss, the lot number on the implant, the exact finding — are the details that are lost first. Where the full note genuinely must wait, a brief operative note goes into the file immediately, and the full note follows the same day.

The test of a good operative note is whether a surgeon who has to reoperate in two years can reconstruct what was done. That requires the findings to be described rather than summarised, the steps to be recorded in enough detail to be followed, and anything unexpected to be stated explicitly.

Thin operative note	Note that would survive a reoperation
Lap chole done. Uneventful. Drain placed. GA.	Laparoscopic cholecystectomy. Four-port technique, 10 mm umbilical, 10 mm epigastric, two 5 mm right subcostal. Findings: thick-walled gall bladder with dense omental adhesions to the fundus; Calot's triangle obscured. Adhesiolysis performed; critical view of safety obtained before clipping. Cystic duct and artery doubly clipped and divided. Gall bladder removed from the liver bed with hook diathermy, retrieved in a bag through the umbilical port. Bile spillage minimal, peritoneal lavage performed. No stones spilled. Liver bed haemostasis confirmed. 16 Fr suction drain to the sub-hepatic space through the right lateral port. Estimated blood loss 40 mL. Ports closed, 10 mm sites closed in sheath. Specimen sent for histopathology, labelled and signed. No complications. Post-operative: oral fluids after four hours, drain removal when output is below 30 mL in twenty-four hours, analgesia as charted.

7.5 Implants and prostheses

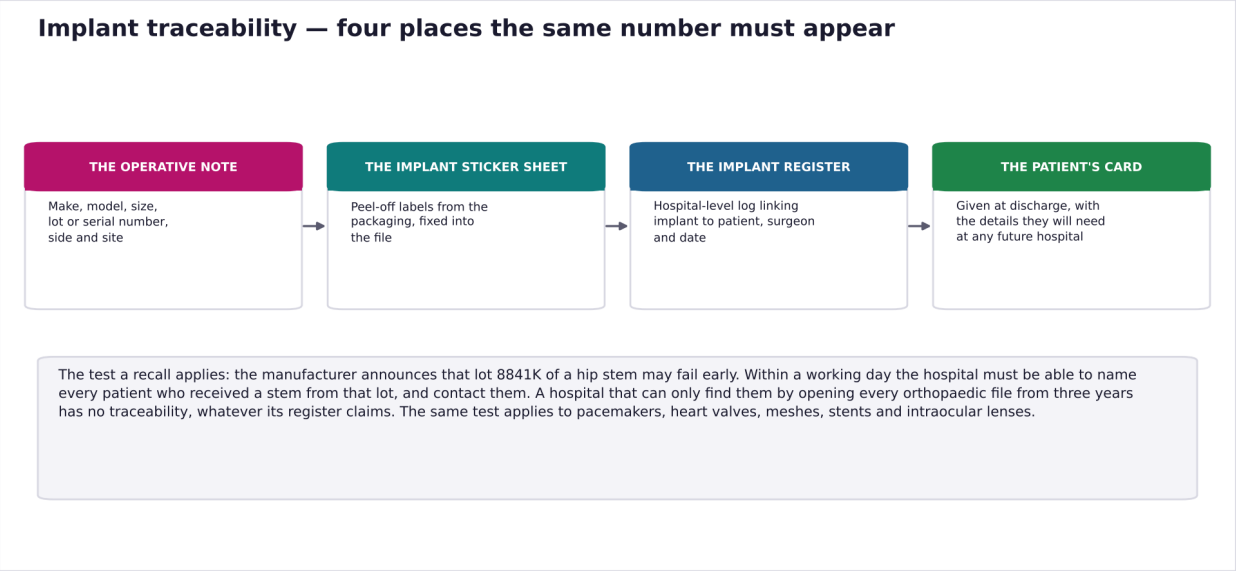


Figure 7.4 — The four places an implant's details must appear.

The test an implant record must pass is the recall test. A manufacturer announces that a lot may fail early. Within a working day the hospital must be able to name every patient who received an implant from that lot and contact them. A hospital that can only find them by opening every file in that specialty has no traceability, whatever its register claims.

- Make, model, size, lot or serial number, and the side and site, recorded in the operative note
- The peel-off labels from the packaging fixed into the record, which is the simplest defence against transcription error
- A hospital-level implant register that links implant to patient, surgeon and date, and is searchable by lot number
- An implant card given to the patient at discharge with the details they will need at any future hospital
- Explants recorded with the same detail, including the reason for removal, because explant data is what identifies a failing device

The same discipline applies to pacemakers and implantable defibrillators, heart valves, stents, meshes, intraocular lenses, orthopaedic prostheses, spinal instrumentation and dental implants.

7.6 Anaesthesia and sedation records

The anaesthesia record is a continuous physiological record and is one of the most scrutinised documents in any case involving an intraoperative event, because it carries the timeline. It must record pre-induction baseline observations, drugs with doses and times, airway management including the grade of view and the number of

attempts, ventilation settings, continuous vital signs at the specified interval, fluids and blood given, blood loss and urine output, and any event with the time and the response.

Sedation outside theatre

Procedural sedation in endoscopy, radiology, cardiac catheterisation, emergency departments and dental suites is held to the same documentation standard as anaesthesia in theatre, and is frequently documented far less well. The record must show the pre-sedation assessment including airway and fasting status, the consent, the drugs and doses, continuous monitoring including oxygen saturation, the sedation level achieved, recovery observations, and the discharge criteria met before the patient left.

> **CAUTION**
> Where a patient is sedated by the same person performing the procedure, the record must show who was monitoring the patient. A record that shows one clinician performing and nobody monitoring is evidence of an unsafe staffing model, not merely a documentation gap.

7.7 The recovery record

The post-anaesthesia care record must show observations at the frequency the policy requires, a recovery score using whichever tool the hospital has adopted, pain assessment and the analgesia given, nausea and vomiting, the state of the airway and oxygen requirement, drains and output, and — the item most often missing — the explicit statement that discharge criteria were met, with the time, and the name of the person to whom the patient was handed over on the ward.

7.8 Procedures outside the operating theatre

Central line insertion, chest drain insertion, lumbar puncture, ascitic and pleural tap, cardioversion, endoscopy, bone marrow aspiration, dialysis catheter insertion and bedside tracheostomy all carry real risk and are all routinely documented in two lines. They require a procedure note to the same standard as an operative note.

Element	Example for a central venous catheter insertion
Indication	Inotropic support required; peripheral access inadequate
Consent	Verbal consent from the patient, explained risks of pneumothorax, bleeding, infection and arrhythmia; witnessed by staff nurse
Operator and supervision	Inserted by registrar under direct consultant supervision
Preparation	Full aseptic precautions, chlorhexidine skin preparation, maximal barrier precautions, ultrasound guidance used
Technique and site	Right internal jugular vein, Seldinger technique, ultrasound-guided, first pass successful
Device	Triple lumen 7 Fr, 20 cm, secured at 16 cm at the skin
Complications	None; no arterial puncture, no arrhythmia during guidewire insertion
Confirmation	Blood aspirated from all three lumens, transduced waveform venous, chest radiograph at 14:20 confirms tip at the cavo-atrial junction and no pneumothorax
Post-procedure	Dressing applied with date; line care per protocol; review daily for ongoing need

7.9 Specimens and the histopathology chain

A specimen leaves the patient and must arrive at a report without any break in identification. The chain is: labelled in theatre with the patient’s identifiers and the site, confirmed aloud at sign out, recorded in the operative note, entered in the theatre specimen register, received and acknowledged by the laboratory, and the report filed in the record and acted on.

- Record in the operative note exactly what was sent and from where, including laterality and the number of containers
- Record who labelled the specimen and who confirmed it
- Record the time it left theatre and the person who transported it
- Track outstanding histopathology reports actively; a malignancy reported on a specimen nobody followed up is one of the most damaging failures a hospital can have
- Record the clinical action when the report arrives, including communication with the patient

> **KEY POINT**
> Every hospital should be able to produce a list of histopathology reports not yet matched to a clinical action. Where that list does not exist, reports are being filed unread, and the hospital will discover it only when a patient presents late with a cancer that was diagnosed and never communicated.

7.10 The surgical safety checklist as a record

The WHO checklist is a clinical control and, once signed, a legal record. Two documentation failures recur. The first is pre-signing — completing the checklist before the checks, which is a false record and is easily proved by comparing it with the anaesthesia timeline. The second is a checklist signed as complete when a step was not performed, typically the team introductions or the anticipated critical events discussion.

> **CAUTION**
> If a checklist step was not performed, record that it was not performed and why. A truthful incomplete checklist is a manageable finding. A complete checklist that contradicts the rest of the record is a credibility problem affecting the whole file.

7.11 Imaging requests and reports

Imaging generates two documentation duties that are frequently neglected: the request must carry enough clinical information for the study to be interpreted, and the report must be acted on and the action recorded.

- The clinical question, written as a question rather than as a diagnosis label
- Relevant history: previous imaging, surgery, implants, devices and known pathology
- Pregnancy status for any ionising study in a woman of child-bearing age, recorded explicitly
- Renal function and allergy history where contrast is planned, with the consent for contrast recorded
- The requesting clinician identified, so the radiologist can discuss an unexpected finding
- The report filed and referred to in the clinical notes with the action taken
- Discrepancies between a provisional and a final report reconciled, with the clinical consequence recorded

> **CAUTION**
> Unexpected significant findings — the lung nodule on an abdominal CT, the vertebral lesion on a spine film taken for trauma — are a recognised source of harm, because the finding is incidental to the reason for the study. The hospital needs a documented route by which such findings are communicated and acknowledged, and the acknowledgement must appear in the record.

7.12 Point-of-care testing

Blood glucose meters, blood gas analysers, haemoglobin devices, urine dipsticks and bedside coagulation devices produce results that drive immediate clinical decisions and are often not documented at all, or are written on a scrap of paper.

- Every point-of-care result is recorded in the patient's record with the value, the time, the device used and the operator
- Quality control records for the device are maintained separately and are part of the hospital's records
- Operators are trained and the competence record is maintained
- Where a point-of-care result drives a treatment decision, the decision and the value are recorded together
- Where a point-of-care result conflicts with a laboratory result, both are recorded and the discrepancy addressed

7.13 Sample rejection and repeat testing

A rejected sample is a clinical event: the result is delayed, the patient is bled again, and sometimes a decision is made without the result. The rejection, its reason and the repeat must be documented, not just absorbed.

Rejection reason	What the record should show
Unlabelled or mislabelled sample	That it was rejected, that it was not relabelled and sent, and that a fresh sample was drawn after positive patient identification
Haemolysed or clotted sample	The rejection, the time, and the repeat collection time, so the delay is visible
Insufficient volume	The repeat, and in a patient with difficult access, the cumulative blood taken
Wrong container or wrong additive	The rejection and the corrective action; a repeated pattern indicates a training or stocking problem
Delayed transport	The collection and receipt times, which is how transport delay is proved

> **KEY POINT**
> A mislabelled sample is never relabelled. It is discarded and recollected, and the event is reported. Relabelling is how a result is attributed to the wrong patient, and the documentation then looks entirely normal.

7.14 Theatre registers and the count

Theatre registers are permanent records and are the index that makes most perioperative traceability possible. The operation register, the anaesthesia register, the implant register, the specimen register and the sterilisation load records together allow a hospital to answer questions that no individual file can answer.

The surgical count — instruments, sponges, needles and sharps — is documented at the intervals the policy requires, with the names of the two people counting. Where a count is incorrect, the record must show the discrepancy, the search conducted, the imaging performed, the outcome, who was informed, and the incident report raised. A count recorded as correct when it was not is a falsified record in the most consequential possible context.

7.15 Devices, drains and lines

Every invasive device should have an insertion record and a removal record, and the interval between them should be visible, because device days drive infection risk and infection surveillance.

Device	Record at insertion	Record at removal
Central venous catheter	Indication, site, operator, aseptic technique, ultrasound use, number of attempts, confirmation of position	Date, time, reason, condition of the site, tip sent for culture where indicated
Urinary catheter	Indication, size, date, operator, and the planned review date	Date, time, reason, and whether the patient voided afterwards
Peripheral cannula	Site, size, date, operator	Date, reason, and the condition of the site using the hospital's phlebitis scale

Device	Record at insertion	Record at removal
Surgical drain	Type, site, date, and what it is draining	Date, total output, and the reason for removal
Endotracheal tube and tracheostomy	Size, depth at the teeth or flange, date, operator, grade of view	Date, time, and the plan for airway support afterwards

7.16 Putting this module to work

1. Take the last operative note you wrote or read and check it against the element list. Could another surgeon reoperate from it?
2. Pick one implant used in your hospital last month and try to trace it from the lot number to the patient. Time how long it takes.
3. Find the last critical result called to your ward and check whether a clinical note records the action taken.
4. Take one recovery record and check whether the discharge criteria are explicitly recorded as met, with a time.
5. Ask whether your hospital has a list of histopathology reports with no recorded clinical action.

Knowledge Check

Work through each item before reading the model answer. These are applied questions: the answer is the standard of documentation your assessor, your medico-legal adviser or the next clinician to read the file would expect.

Q1. **A potassium of 7.1 mmol/L is telephoned to the ward at 23:40, read back and logged by the laboratory. The patient is treated and recovers. The clinical notes contain nothing about it until the morning round. What is the problem?**

Answer: The hospital can prove the laboratory did its job and cannot prove that anyone acted on the result in a timely way. The treatment given is undocumented, so there is no record of what was given, when, or what the subsequent potassium was. The clinician receiving a critical value must make a clinical entry at the time recording the value, the time received, the action taken and the follow-up arranged. The gap between the laboratory log and the clinical note is where these cases are lost.

Q2. **A surgeon writes operative notes for a five-case list at the end of the day. What specifically is likely to be lost, and what is the correct practice?**

Answer: Estimated blood loss, implant lot numbers, the exact anatomical findings, which drains were placed and where, and whether anything unexpected occurred. Memory merges similar cases. The correct practice is to write the note immediately after each case, before leaving the theatre suite. Where that is genuinely impossible, a brief operative note goes in the file at once with the essential findings and implants, and the full note follows the same day.

Q3. **A manufacturer recalls a lot of hip stems. Your hospital's implant register records patient name, surgeon and date, but not the lot number. What can the hospital do and what should it have had?**

Answer: It can only identify candidate patients by specialty and date range and then open every file to look for implant stickers, which takes days and will miss files that are out on loan or incomplete. It should have had the lot or serial number in the register, searchable, with the packaging labels also fixed into each record, the details in the operative note, and an implant card issued to each patient. Traceability must work in both directions: lot to patient, and patient to every implant.

Q4. **An endoscopy unit sedates patients and discharges them when they look awake. There is no sedation record and no documented discharge criteria. What are the risks?**

Answer: Clinically, patients are discharged while still sedated, with risks of airway compromise, falls and road traffic accidents. In documentation terms, the hospital cannot show the pre-sedation assessment, the drugs and doses, that monitoring occurred, that recovery was adequate, or who escorted the patient home. Procedural sedation outside theatre is held to the same standard as anaesthesia in theatre, and this unit meets none of it.

Q5. **A surgical safety checklist is fully completed and signed, but the anaesthesia record shows induction at 09:05 and the checklist is timed 09:20. What does this indicate?**

Answer: That the sign-in, which must precede induction, was recorded after it — so the checklist was completed retrospectively. This undermines the checklist as a control and as a record, and it casts doubt on the rest of the perioperative documentation. The correct response is to record each phase at the time it occurs, and where a phase was missed, to record that it was missed and why rather than back-filling it.

Q6. **A specimen sent from theatre is reported three weeks later as malignant. The report is filed in the record. The patient is not contacted and presents eight months later with advanced disease. Which documentation controls failed?**

Answer: The outstanding-report tracking failed: there was no active list of histopathology reports awaiting a clinical action. The result acknowledgement failed: no clinician recorded reading the report. The communication documentation failed: no note of informing the patient, because no one did. The controls needed are a specimen register tracked to report, a named clinician responsible for every pending report, mandatory acknowledgement of significant results, and an entry recording what the patient was told and when.

ASSESSMENT BANK
A fuller bank of 50 scored questions for this module is supplied separately as a JSON question bank for loading into your LMS.